Project Report

Streamlining IT Procurement:
Automating Standard Laptop Orders with Flow Designer

Team ID: _____

Team Members

Team Member 1: Project lead

Team Member 2: ServiceNow developer

Team Member 3: Business analyst

Team Member 4: IT procurement coordinator

Team Member 5: Tester

Contents

1. Introduction
- 1.1 Project Overview
- 1.2 Purpose

2. Ideation Phase
- 2.1 Problem Statement
- 2.2 Empathy Map Canvas
- 2.3 Brainstorming

3. Requirement Analysis
- 3.1 Customer Journey Map
- 3.2 Solution Requirements
- 3.3 Data Flow Diagram
- 3.4 Technology Stack

4. Project Design
- 4.1 Problem Solution Fit
- 4.2 Proposed Solution
- 4.3 Solution Architecture

5. Project Planning & Scheduling
- 5.1 Project Planning

6. Functional and Performance Testing
- 6.1 Performance Testing

7. Results
- 7.1 Output Screenshots

8. Advantages & Disadvantages

9. Conclusion

10. Future Scope

11. Appendix
- Source Code
- Dataset Link
- GitHub & Project Demo Link

1. INTRODUCTION

1.1 Project Overview

In many organizations, getting a standard laptop still relies on email threads, manual sign-offs and purchase orders typed in by hand. This project moves that process into ServiceNow using Flow Designer. The employee fills in a single catalog request. The flow verifies eligibility, approves straight away for pre-approved roles or sends the manager a one-click approval, checks inventory, raises a purchase order if the model is out of stock, creates the asset record and keeps the employee informed at every stage.

1.2 Purpose

The project aims to put approved devices in employees' hands sooner, take repetitive procurement tasks off IT staff and give everyone a clear view of where each request stands. The expected outcomes are shorter turnaround time, fewer manual steps and fewer mistakes.

2. IDEATION PHASE

2.1 Problem Statement

How might we automate standard laptop orders in ServiceNow with Flow Designer so that employees get approved devices sooner and IT spends less time on repetitive procurement tasks?

PS	I am	I am trying to	But	Because	Which makes me feel
PS-1	A newly hired employee	Receive a laptop before my joining date	Every request needs several manual approvals and reminders	Requests travel by email and each step waits for a person	Worried and unable to start work
PS-2	A department manager	Approve my team's requests without delay	Requests arrive with incomplete information and repeated chasers	No rule exists to pre-approve eligible roles	Overwhelmed and unsure about budget
PS-3	An IT procurement coordinator	Fulfil orders correctly	I retype details, check stock manually and follow up on status	The steps are not linked in one automated flow	Frustrated and overworked

2.2 Empathy Map Canvas

Empathy maps were prepared for the requesting employee (Meera) and the IT procurement coordinator (Kiran).

	Meera, requesting employee	Kiran, IT procurement coordinator
Says	"Where is my laptop?" "Who can give me a status update?"	"I keep retyping the same information." "Is this model available in stock?"
Thinks	"A standard laptop should not need this many approvals."	"Nearly all of these requests look the same."
Does	Messages IT and her manager for updates and keeps following up	Looks up inventory manually, chases approvals and raises orders by hand
Feels	Worried, frustrated, unsure whether the request is progressing	Overworked by repeated tasks and constant status queries
Pains	No visibility, long waits, entering the same details again	Manual hand-offs, status queries, mistakes from retyping
Gains	An easy request, automatic approval, a laptop ready on day one	Routine steps automated, time for exceptions, reliable records

2.3 Brainstorming

The team of five brainstormed ideas, sorted them into five themes and plotted them on an importance and feasibility grid.

Theme	Ideas
Make it simple to request the correct laptop	Standard catalog item with pre-filled specifications; auto-fill from the user record; eligibility check and duplicate blocking
Approve fast without manual follow-up	No approval needed for pre-approved roles; one-click manager approval; automatic escalation; budget check by cost center
Fulfil automatically from stock or supplier	Inventory check and reservation; automatically created purchase order; vendor assignment by region or price
Keep everyone updated	Notifications at every stage; status tracker on the portal
Close the loop and measure outcomes	Asset record created automatically on delivery; turnaround and savings dashboard

Prioritization: Ideas rated high on both importance and feasibility (no approval for pre-approved roles, the standard catalog item, inventory check and reservation, stage notifications) are built first. The budget check, automatic purchase order and escalation follow. The asset record, vendor assignment and dashboard are planned for a later phase.

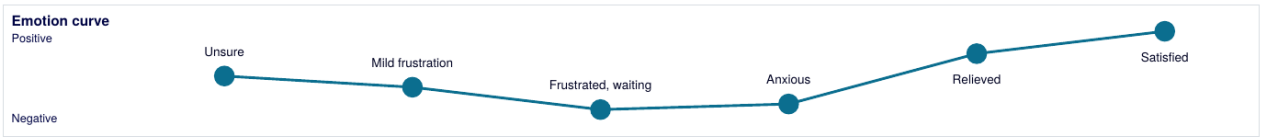
3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey map follows Meera through six stages, from realizing she needs a laptop to closing the request. The lowest point of her experience is the approval and fulfilment stage, where she waits with no visibility.

	1. Need arises	2. Submit request	3. Approval	4. Fulfilment	5. Delivery and setup	6. Close and review
Customer actions	Employee joins or needs a device replacement and finds the laptop request option	Opens the service catalog, picks the standard laptop and submits the request	Manager receives the approval task and approves or rejects it	Stock is checked; a purchase order is raised if the laptop is not in stock	Receives the laptop, signs for it and logs in	Confirms receipt and gives feedback on the process
Touchpoints	Employee portal, onboarding checklist	Service catalog form	Email and mobile approval	Inventory records, vendor order	Delivery notice, asset record	Satisfaction survey, request status page
What they think	"Where do I order this?"	"Why is so much information needed?"	"Why is my request waiting?"	"Is anyone working on it?"	"Finally, can I get started?"	"That was easier than before."
Pain points	No single place to start; unclear eligibility	Details re-entered that IT already has	Manual approvals, repeated reminders	Manual stock checks, delayed orders	Asset not recorded or assigned on time	Feedback rarely collected
Opportunities with Flow Designer	Single standard laptop catalog item with eligibility check	Pre-fill details from user record; block duplicates	Auto-approve eligible roles; one-click approval; auto-escalate	Auto stock check and purchase order creation	Auto-create asset record; notify at each stage	Auto survey and turnaround dashboard

Emotional Journey



Key insight: the lowest point is the approval and fulfilment stage, where the employee waits with no visibility. Automating approvals, stock checks, purchase orders and status notifications in Flow Designer removes the manual hand-offs that cause the delay.

Figure 1: Customer journey map and emotional journey

3.2 Solution Requirements

No.	Functional requirement	Sub requirements
FR-1	Laptop request	Standard catalog item; pre-filled form; eligibility and duplicate validation
FR-2	Approval management	Auto-approval for pre-approved roles; manager approval by email or mobile; budget check; escalation after 48 hours
FR-3	Inventory and fulfilment	Stock check and reservation; automatic purchase order; vendor assignment
FR-4	Notifications and tracking	Notifications at each stage; status shown on the portal
FR-5	Asset management	Asset record created and assigned on delivery
FR-6	Reporting	Turnaround time dashboard; report on automated requests and savings

No.	Non-functional requirement	Description
NFR-1	Usability	A standard laptop can be ordered in a few steps from a single catalog item
NFR-2	Security	Role-based access control; request and asset data visible only to authorized users
NFR-3	Reliability	Failed steps are logged and sent to the coordinator; no request is lost
NFR-4	Performance	Routine flow actions finish within seconds with no manual delay
NFR-5	Availability	Matches the availability of the ServiceNow instance
NFR-6	Scalability	New models, roles and vendors are added through configuration

3.3 Data Flow Diagram

The diagram shows how data travels from the employee through approval, stock check, purchase order and delivery, and which data store each process reads or writes.

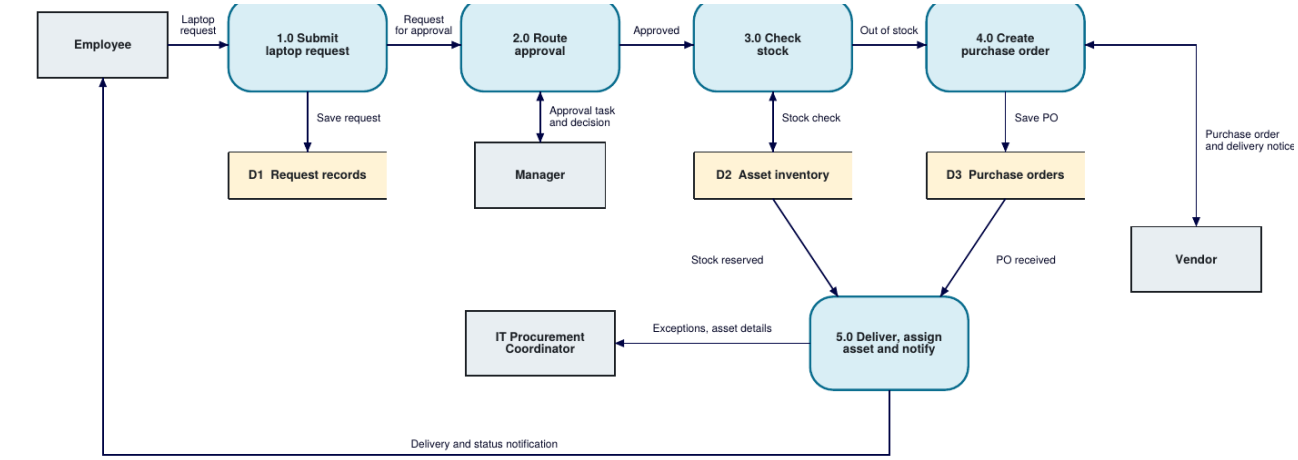


Figure 2: Data flow diagram

3.4 Technology Stack

The solution runs on the ServiceNow platform, with Flow Designer as the automation layer.

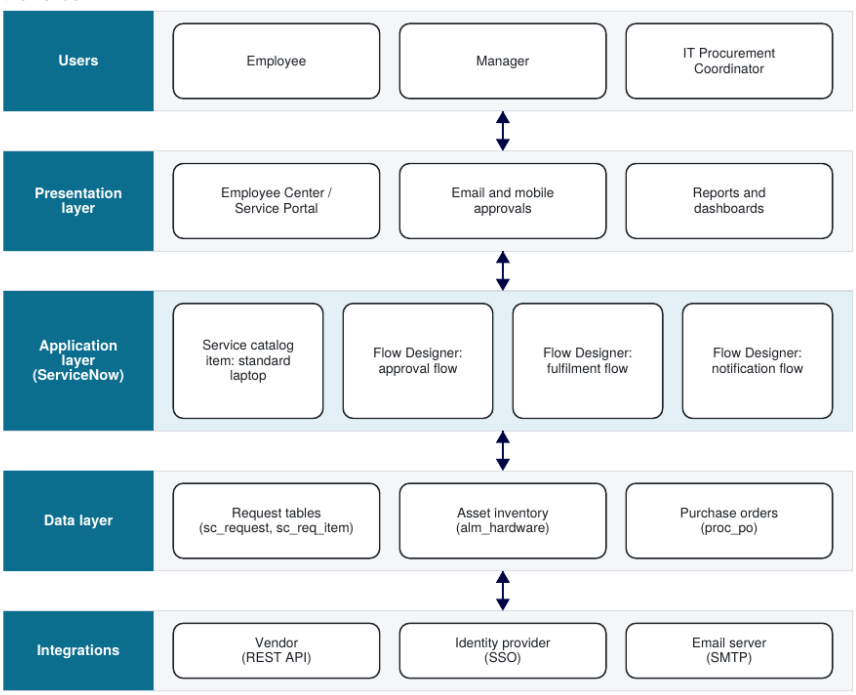


Figure 3: Technical architecture and stack

Component	Technology
User interface	ServiceNow Employee Center / Service Portal
Service catalog	Service Catalog with catalog item and variables
Workflow automation	Flow Designer, subflows and actions
Database	ServiceNow tables (sc_request, sc_req_item, alm_hardware, proc_po)
Notifications	ServiceNow Notifications by email and mobile
External integration	REST Message, IntegrationHub
Reporting	ServiceNow Reports and Performance Analytics

4. PROJECT DESIGN

4.1 Problem Solution Fit

Element	Summary
Customer segments	New and current employees, approving managers and IT procurement coordinators
Jobs / problems	Receive an approved laptop quickly; approve without delay; fulfil orders correctly. Problems: manual approvals, no status visibility, repeated data entry
Triggers	New hire, device refresh cycle, lost or damaged device, team expansion
Emotions	Before: worried, frustrated. After: relieved, confident
Available solutions	Email and spreadsheets; generic catalog with manual approvals; manual stock checks
Constraints	Budget limits, approval and security policies, stock and vendor lead times, limited IT capacity
Root cause	Manual hand-offs, no automated rules for standard requests, no single view of status
Our solution	Flow Designer automation with one catalog item, notifications and a dashboard

4.2 Proposed Solution

Parameter	Description
Problem statement	Manual hand-offs slow down laptop orders and consume IT time on repetitive tasks
Idea / solution	Automate the order in ServiceNow using Flow Designer, from the request to the asset record and notifications
Novelty	Request, approval, inventory, purchasing and asset records joined in one low-code flow, with rules that can be changed through configuration
Social impact / satisfaction	Quicker devices, visible status and less routine work for managers and IT
Business model	Internal efficiency solution on the existing ServiceNow platform; value comes from shorter turnaround and less manual effort
Scalability	Copes with growing volumes and can be reused for other standard hardware

4.3 Solution Architecture

The architecture shows the employee, manager and coordinator working inside the ServiceNow platform, where the service catalog triggers the Flow Designer flow. The flow reads and writes the platform tables and connects to the identity provider, email server and vendor.

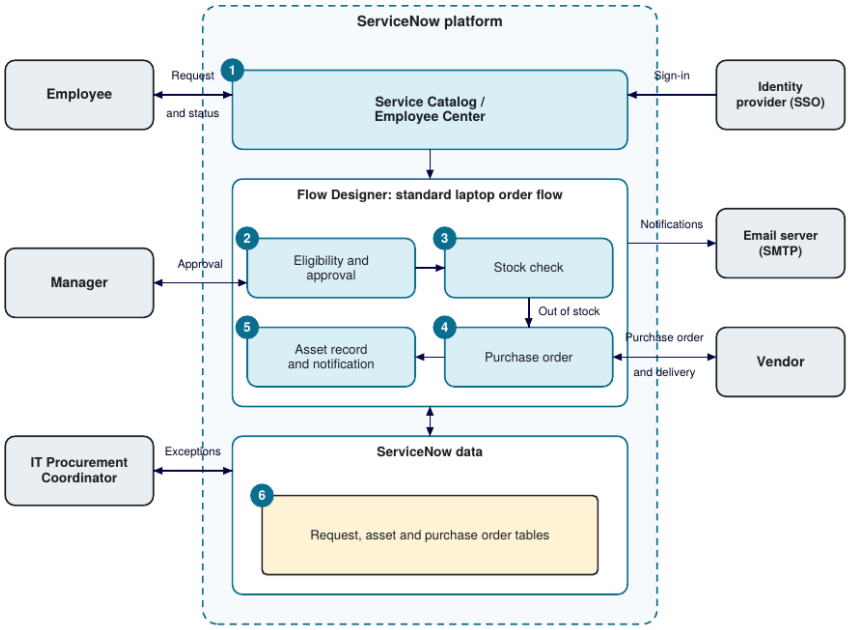


Figure 4: Solution architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The 11 user stories are divided into three sprints of 6 days each and estimated in story points. The plan totals 51 story points, an average velocity of about 17 points per sprint or 2.8 points per day.

Sprint	Stories	Story points	Dates (planned)
Sprint-1	USN-10 platform configuration, USN-1 laptop request, USN-5 auto-approval	14	17 Feb to 22 Feb 2025
Sprint-2	USN-4 manager approval, USN-3 notifications, USN-6 stock check, USN-7 purchase order	20	24 Feb to 01 Mar 2025
Sprint-3	USN-2 status tracking, USN-8 asset record, USN-9 dashboard, USN-11 testing and release	17	03 Mar to 08 Mar 2025

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional tests verify each step of the flow, while performance tests measure success rate, execution time and load. User acceptance testing was carried out with the employee, manager and coordinator roles.

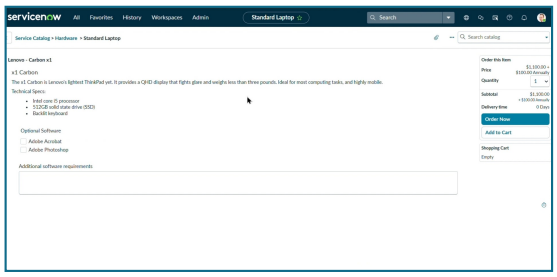
ID	Scenario	Expected result	Result
FT-01	Request form validation	Correct requests accepted; errors shown for invalid ones	Pass
FT-02	Quantity and duplicate validation	Excess quantity and repeat orders rejected	Pass
FT-03	Approval and fulfilment flow	Right route for each approval and stock outcome	Pass
FT-04	Vendor and email integration	Vendor responds; email received	Pass
PT-01	Flow execution time	Under 5 seconds (average 3.8 seconds)	Pass
PT-02	60 simultaneous requests	No visible slowdown (97% success)	Pass
PT-03	120 bulk requests	Runs without crashing (116 of 120 completed, 4 retried)	Pass

UAT summary: all 6 test cases passed. 3 bugs were logged: 2 were fixed and closed, and 1 (the approve button on small phone screens) is still in progress.

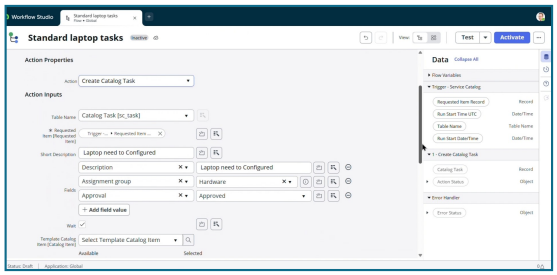
7. RESULTS

7.1 Output Screenshots

Add screenshots of the working solution below.



Add screenshot
Manager approval email



Add screenshot
Turnaround and savings dashboard

8. ADVANTAGES & DISADVANTAGES

Advantages

- Quicker laptop delivery, as routine approvals and stock checks run on their own.
- Less manual effort and fewer mistakes for managers and IT staff.
- Employees get clear status and notifications.
- Low-code, so rules and vendors can be updated through configuration.
- Built on the ServiceNow platform already in use, so no new tools need to be purchased.

Disadvantages

- Relies on the ServiceNow platform and its uptime.
- Needs accurate asset inventory and user data to perform well.
- Covers standard laptops only; non-standard requests still need manual handling.
- The vendor integration needs upkeep if the vendor changes its endpoint.

9. CONCLUSION

The project demonstrates that a standard laptop order can be automated from start to finish in ServiceNow using Flow Designer. The solution tackles the problems found with the people involved: waiting for manual approvals, entering the same data repeatedly and having no view of status. In testing, the flow ran reliably and quickly, and user acceptance testing passed every planned case. The one open issue is the approve button on small phone screens.

10. FUTURE SCOPE

- Extend the flow to other standard hardware such as monitors, phones and accessories.
- Automatically compare vendor prices before a purchase order is raised.
- Add a mobile-friendly approval screen and chatbot-based requests.
- Predict stock needs from hiring plans and refresh cycles.

11. APPENDIX

Source Code: The solution is built from low-code Flow Designer flows, subflows and a catalog item instead of written code. Add an update set export or flow screenshots here if required.

Dataset Link: Not applicable. The solution uses ServiceNow tables filled with sample records. [add link if required]

GitHub & Project Demo Link: [add GitHub link here] and [add demo link here]

Note: the dates, test results, counts and figures in this report are sample entries carried over from the earlier project documents. Replace them with your real results before you submit.